MONTECARLO

1) Genero X variable aleatoria según $p(x) = x^{-2,7}$ a partir de $r \in [0,1]$

i. Comprobar normalización:

$$\int_1^\infty p(x)\,dx = \int_1^\infty x^{-2,7}\,dx = -\frac{x^{-1,7}}{1,7} = -\frac{1}{1,7} = -\frac{1}{1,7}$$: hay que normalizar

(1) por qué estos límites

$$p(x) = 1,7 \cdot x^{-2,7}$$

$$p_0(r) = 1 = \left|\frac{dx}{dr}\right| \cdot p(x) = \frac{dx}{dr} \cdot 1,7 \cdot x^{-2,7} \Rightarrow \int_0^r dr = \int_1^x 1,7 \cdot x^{-2,7} \Rightarrow r = \frac{1,7}{-1,7} \cdot x^{-1,7}\Big]_1^x = -1\left(x^{-1,7} - 1\right) =$$

$$= 1 - x^{-1,7} \Rightarrow 1 - r = x^{-1,7} \Rightarrow x^{1,7} = \frac{1}{1-r} \Rightarrow x = \sqrt[1,7]{\frac{1}{1-r}}$$

2) Dist. de probabilidad:

$x(r) = 10^{3r}$ ¿p(x)? $r \in [0,1]$

$$x(r) = 10^{3r} \Rightarrow \log x = 3r \Rightarrow r = 1/3 \log x = \int_1^x p(x)\,dx$$

$$\log_{10^x}\Big|' = \frac{1}{x} \cdot \frac{1}{\ln 10} = p(x)$$

- Comprobar:

$$p(x) = \frac{1}{x \cdot \ln 10} \rightarrow \int_1^{1000} dx\, p(x) = \frac{1}{\ln 10}\left[\ln x\right]_1^{1000} = 1/\ln 10 \cdot \ln 1000$$

✓ $p(x) = \frac{1}{x} \cdot \frac{1}{\ln 1000}$ $\rightarrow \int_1^{1000} \frac{1}{k}\,dx \cdot \frac{1}{\ln 1000} = \frac{1}{\ln 1000} \cdot \ln 1000$

$$\int_0^r r = \int_1^x p(x) \cdot dx \Rightarrow r = \frac{1}{\ln 1000} \cdot \int_1^x \frac{1}{k}\,dx = \frac{1}{\ln 1000} \cdot \ln x \Rightarrow$$

$$\Rightarrow r \cdot \ln 1000 = \ln x \Rightarrow r \ln \cdot 10^3 = \ln x \Rightarrow \ln 10^{3r} = \ln x \Rightarrow 10^{3r} = x$$

3) X con $p(x) = 10^{-2x}$ $x \in [1,3]$ a partir de $r \in [0,1]$ $\qquad 10^{-2x}\Big|' = 10^{-2x} \ln 10 (-2)$

i. Normalización

$$\int_1^3 p(x)\,dx = \int_{-1}^3 10^{-2x}\,dx = \frac{10^{-2x}}{-2 \cdot \ln(10)}\Big|_{-1}^3 = \frac{1}{-2\ln 10}\left[10^{-6} - 10^2\right] =$$

$$= \frac{99,999999}{2 \ln 10} \sim 5/\ln 10$$

$$\left\{ p(x) = \frac{1}{k} \cdot 10^{-2x} \right| \checkmark = \frac{2\ln 10}{10^2 - 10^{-6}} \cdot 10^{-2x}$$

$$\int_{-1}^3 1/k \cdot 10^{-2x} = \frac{1}{k}\left[\frac{10^{-2x}}{-2\ln 10}\right]_{-1}^3 = \frac{2\ln 10}{99,999999} \checkmark \cdot \frac{10^{-6} - 10^2}{2\ln 10} = 1$$

$$P_0(r) = 1 = \frac{dx}{dr}\cdot p(x) \Rightarrow \int_0^r dr = r = \int_{-1}^{x} \left(\frac{2\cdot \ln 10}{10^2 - 10^{-6}}\right) 10^{-2x}\,dx = 1/k \cdot \left[\frac{10^{-2x}}{-2\ln x}\right]_{-1}^{x} =$$

$$= -\frac{10^{-2x}}{10^2 - 10^{-6}}\Big|_{-1}^{x} = \frac{10^2 - 10^{-2x}}{10^2 - 10^{-6}} = \frac{10^2(1 - 10^{-2x-2})}{10^2(1 - 10^{-8})} = \frac{1 - 10^{-2x-2}}{1 - 10^{-8}} \Rightarrow 1 - 10^{-2x-2} = (1 - 10^{-8})r \Rightarrow$$

$$\Rightarrow 1 + (10^{-8} - 1)r = 10^{-2x-2} \Rightarrow -2x - 2 = \log[1 + (10^{-8} - 1)r] \Rightarrow x + 1 = -1/2 \log[\ ] \Rightarrow$$

$$\Rightarrow x = -1/2 \log[\ ] - 1$$

iii Comprobación:

$r = 0 \rightarrow x = -1/2 \log 1 - 1 = -1$

$r = 1 \rightarrow x = -1/2 \cdot \log[1 + (10^{-8} - 1)\cdot 1] - 1 = -1/2 \log 10^{-8} - 1 =$

$= -1/2\,(-8) - 1 = 3$

4) 2 variables aleatorias

$x \rightarrow p_1(x)$

$y \rightarrow p_2(x)$

¿ pdf de $z = x/y$?

$g(z) = \int dy \cdot x \cdot p(z\cdot y, y)$

↳ Objetivo: $g(z)\,dz$ ← probabilidad de obtener puntos $z = x/y$ y $x/y = z + dz$

$z = x/y \Rightarrow x = z\cdot y$

$x = x(z, y) = z\cdot y$

$x = zy + y\,dz$

$\frac{dx}{dz} = y \Rightarrow dx = y\,dz \rightarrow dx = dx(z)\cdot dz = y\cdot dz$

$$g(z)\,dz = \int dx\,dy\, p(x,y) \cdot \int dy \left|\frac{dx}{dz}\right| p(x(z), y) =$$

(¿de dónde sale)

$$= \int_{-\infty}^{+\infty} dy \int_{yz}^{yz+y\,dz} dx\, p(x,y) = \int_{-\infty}^{+\infty} dy \cdot y \cdot p(z\cdot y, y)\,dz \Rightarrow$$

$$\Rightarrow g(z) = \int_{-\infty}^{+\infty} y\cdot dy\, p(zy, y) = \int_{-\infty}^{+\infty} y\,dy\, p_1(zy)\cdot p_2(y).$$

5) PROBLEMA DE EXAMEN

$$p(x,y) = \frac{1}{\pi} \cdot e^{-x^2-y^2}$$

a) A partir de $r \in [0,1]$ generar pares (x,y) con $p(x,y)$

$$p(x,y) = \frac{1}{\pi}\left[e^{-x^2} \cdot e^{-y^2} \right] = p_1(x) \cdot p_2(y)$$

Se trata de 2 variables independientes, separables

i. $p(x) \cdot \int_{-\infty}^{\infty} \frac{e^{-x^2}}{\sqrt{\pi}} \cdot dx = \frac{1}{\sqrt{\pi}} \cdot \sqrt{\pi} = 1$ ✓

$$1 = \frac{dx}{dr}\, p(x) \Rightarrow \int_0^r dr = r = \int_{-\infty}^{x} e^{-x^2} dx$$

$(x,y) \to (r,\theta)$

$r = \sqrt{x^2+y^2}$ $\qquad p(r) = \frac{1}{\pi} \cdot e^{(-1)(x^2+y^2)} = 1/\pi \cdot e^{-r^2}$

$\theta = \arctg(y/x)$

$x = r\cos\theta$ $\qquad q(r,\theta) = p(x,y)|J| = r \cdot 1/\pi\, e^{-r^2}$

$y = r \cdot \sin\theta$

$$J = \begin{vmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{vmatrix} = \begin{vmatrix} \partial f_1/\partial x_1 & \partial f_1/\partial x_2 \\ \partial f_2/\partial x_1 & \partial f_2/\partial x_2 \end{vmatrix} = r$$

$$1 = \int d\theta \cdot dr\, p(\theta, r) = \int_0^{2\pi} d\theta \int_0^{\infty} dr\, r \cdot \frac{1}{\pi} e^{-r^2} = 2\pi \cdot \left(-\frac{1}{2}\right) \frac{e^{-r^2}}{\pi} \Big|_0^{\infty} = (-1)\left[0 - e^0\right] = +1$$

Comprobar la normalización

$$q(r,\theta) = r \cdot 1/\pi\, e^{-r^2} = \frac{2}{2} \cdot \frac{1}{\pi} \cdot r e^{-r^2} = \frac{1}{2\pi} \cdot 2r e^{-r^2} d = q(\theta)\, q_2(r)$$

↳ Funciones separables: generamos cada 1 de forma independiente

θ: $1 = \int_0^{2\pi} d\theta \cdot p(\theta) = \frac{1}{2\pi} \cdot 2\pi = 1$

r: $1 = \int_0^{\infty} 2r e^{-r^2} dr = 2 \cdot -\frac{1}{2} e^{-r^2} \Big|_0^{\infty} = +1$

$$1 = \frac{d\theta}{dr'} \cdot p(\theta) \Rightarrow \int_0^{r'} dr' = r' = \int_0^{\theta} \frac{1}{2\pi} \cdot d\theta \Rightarrow 2\pi r' = \theta$$

$\frac{d\theta}{dr'} \in [0,1]$

$$1 = \frac{dr}{dr'} p(r) \Rightarrow r' = \int_0^r dr \cdot 2r e^{-r^2} = 2\left(-\frac{1}{2}\right) e^{-r^2} \Big|_0^r = (-1)\left(e^{-r^2} - 1\right) = 1$$

$$\Rightarrow -r' + 1 = e^{-r^2} \Rightarrow -r^2 = \ln(1-r') \Rightarrow r^2 = -\ln(1-r') = \ln\left(\frac{1}{1-r'}\right) \Rightarrow$$

$\Rightarrow r = \sqrt{\ln\left(\frac{1}{1-r'}\right)}$

$x = r\cos\theta = \sqrt{\ln\left(\frac{1}{1-r'}\right)} \cdot \cos(2\pi r')$

$y = r\sin\theta = \sqrt{\ln\left(\frac{1}{1-r'}\right)} \cdot \sin(2\pi r')$

b) $z = y/x = \tan\theta = \tan(2\pi r')$

$z = \tan(2\pi r') \Rightarrow 2\pi r' = \operatorname{arctg} z \Rightarrow r' = 1/2\pi \operatorname{arctg} z$

$1 = \frac{dz}{dr'} p(z) \Rightarrow \int_0^{r'} dr' = r' = \int_{-\infty}^{\infty} dz\, p(z)$

$\vdots$

$\operatorname{arctg} z|' = \frac{1}{1+z^2}$

$p(z) = \frac{1}{\pi}\frac{1}{1+z^2} \rightsquigarrow \int_{-\infty}^{\infty} \frac{1}{\pi}\frac{1}{1+z^2} dz = \frac{\operatorname{arctg} z}{\pi}\Big|_{-\infty}^{\infty} = \frac{\pi/2 - (-\pi/2)}{\pi} = \frac{\pi}{\pi}$

$\int_0^{r'} dr' = r' = \int_{-\infty}^{z} \frac{1}{1+z^2}\frac{dz}{\pi} = \frac{1}{\pi}(\operatorname{arctg} z - \pi/2) \Rightarrow$

$\Rightarrow \pi \cdot r' + \pi/2 = \operatorname{arctg} z \Rightarrow z = \tan(2\pi r + \pi)$

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